Total No. of Pages: 3 *Roll No.....*

VITH SEMESTER **B.Tech.(Computer Engineering)**

END SEMESTEREXAMINATION **(May 2018)**

CO 304 **ARTIFICIAL INTELLIGENCE**

Paper Code

Time: 3 Hours **Max. Marks: 50**

Note: Attempt total 5 questions. Q1 is compulsory

Q1. Answer following to the point.

a. Differentiate between solution steps recoverable and non recoverable? Illustrate with examples.
b. Explain the concept of knowledge acquisition and Inferential procedure in knowledge representation scheme.
c. State Bayes theorem and support with example for statistical reasoning.
d. What are the different techniques for non monotonic reasoning?
e. Explain rule based expert system

(5 X 2)

Q2.(a) Consider the graph in *Figure 1*, the node A is the start node and nodes J, G and R are goal nodes. The tree is being searched by DFS, BFS and DFID algorithm (searching left to right) Write the sequence of nodes inspected bythem till termination. Mention and show each and every step involved.

(b) As you get close to graduating MIT, you decide to do some career planning. *Figure 2* at end shows a graph of your options where the start node is M = MIT and your goal node is R = Retire, with a bunch of options in between. Your graph includes edge distances that represent, roughly, the "cost of transition" between these careers (don't think too hard about what this means). You also have heuristic node-to-goal distances which represent your preconceptions about how many more years you have to work until you retire. For example, you think it will take 25 years to go from MIT (M) to retirement (R), 30 years from Grad School (B), but only 2 years from Entrepreneur (E). Mention A* algorithm and Apply A* to find a respective career path after MIT till retirement.

A = Wall Street | B = Grad School | C = Professor | D = Government | E = Entrepreneur

[6,4]

Q3. (a) Draw the Grundy Game Tree with seven coins. Explain using this game tree how player maximize and minimize decide their move

(b) For the graph in *Figure 3* find the max value at the root node by applying mini max search. Also do alpha beta pruning. Choose and highlight the best strategy for the graph.

[5,5]

Q4. Consider the following sentences:

a. Shiva, Gopal and Madhu are people.
b. Shiva likes all kind of food..
c. Any thing any one eats and is not killed is a food.
d. If you are killed you are not alive.
e. Madhu eats everything Gopal eats.
f. Gopal eats peanuts and is still alive.
g. Apple is a food

a. Represent theses sentences as wff.
b. Using backward chaining prove that (i) Shiva eats apple and (ii) shiva like peanut.
c. Convert them to clause form indicating the law used.
d. Explaining steps of resolution to prove that Madhu eats peanut.

[2,2,3,3]

Q5. Construct the semantic net representations for the following text:

All Moving vehicles have fuel system and engine. All moving vehicles are ground or water vehicles. Car, Scooter, Truck are ground Vehicles which have breaks and water vehicles have navigation system. Scooter is two wheeler, car is four wheeler and truck is large four wheeler. All cars have air condition and safety belts. Maruti Swift is a car Swift car has 10 years free service of spare parts. Dl-12c 7788 is swift car having black color

a. Illustrate the steps of infer procedure to show that above car has safety system
b. Construct the semantic frame representation and give prolog implementation.

[5,5]

Q6. (a)) What are the steps to build Expert system such as Medical diagnostic system.

(b)What are the contents of knowledge Base in Expert system.

(c) For following fuzzy sets, find their Union, Intersection and also find union of their complements sets

$$A = \left\{\frac{1}{2.0} + \frac{0.65}{4.0} + \frac{0.5}{6.0} + \frac{0.35}{8.0} + \frac{0}{10.0}\right\}$$

$$B = \left\{\frac{0}{2.0} + \frac{0.35}{4.0} + \frac{0.5}{6.0} + \frac{0.65}{8.0} + \frac{1}{10.0}\right\}.$$

[4,3,3]

Q7. Write short notes on following (Any two)

(a) Conceptual Dependency
(b) Different phases of Natural Languge processing
(c) Learning using Neural Network or Genetic algorithm.
(d) Forward reasoning inference and Backward reasoning inference in Prolog

[5,5]

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[5,5]

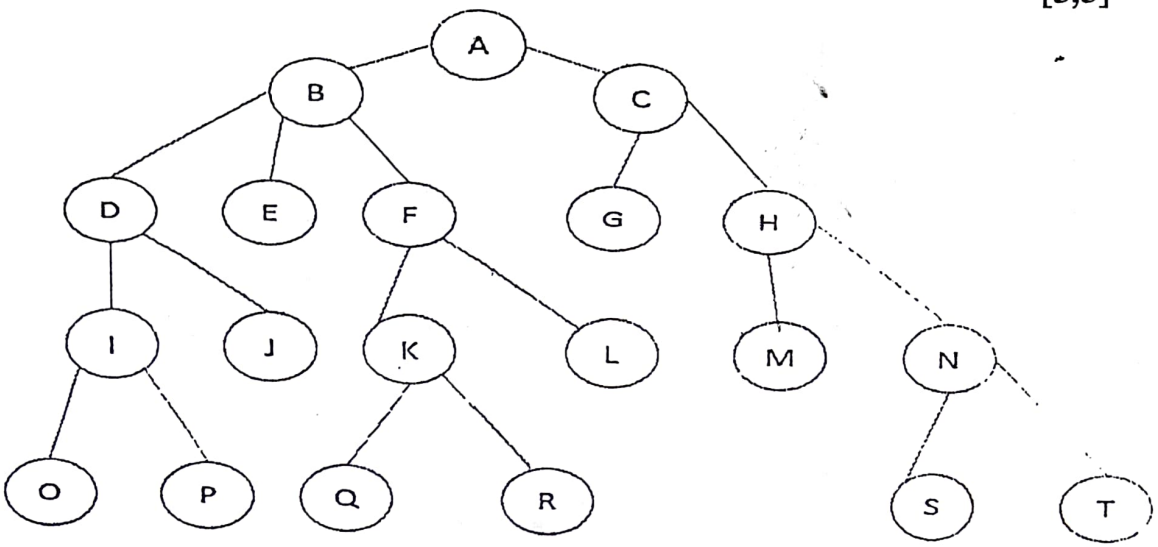

Figure 1

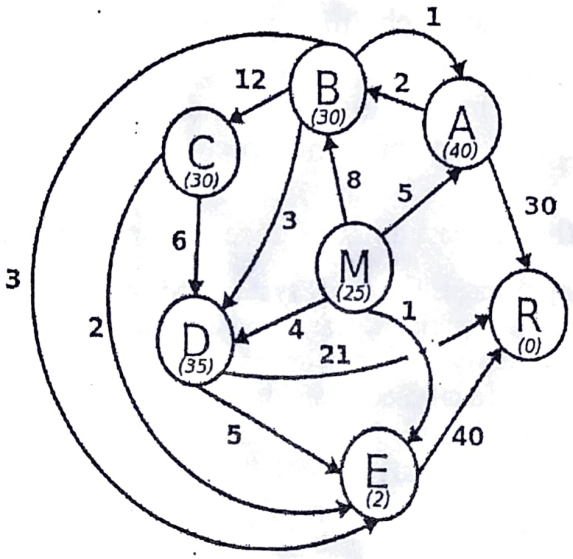

Figure 2.

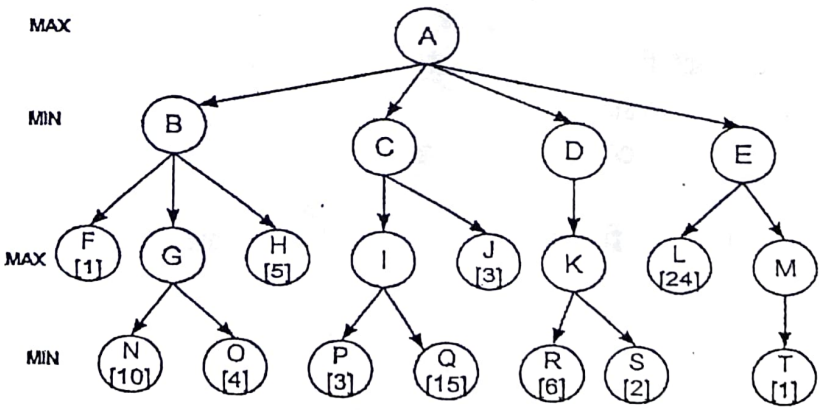

Figure 3.